## Interactive / complex / 1

IC 1
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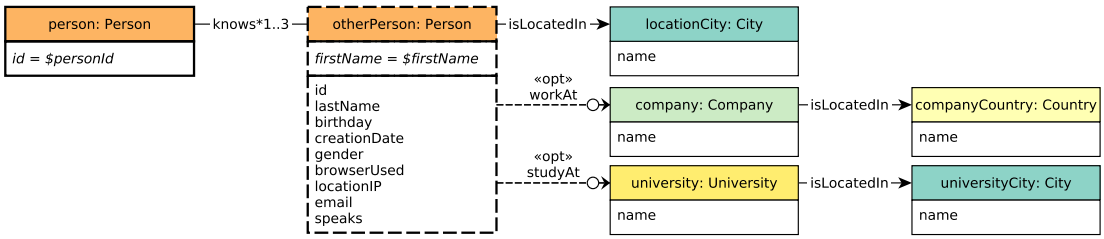
| | |
|---|---|
| query | Interactive / complex / 1 |
| title | Transitive friends with certain name |
| pattern |   |
| desc. | Given a start Person, find Persons with a given first name (`firstName`) that the start Person is connected to (excluding start Person) by at most 3 steps via the knows relationships. Return Persons, including the distance (1..3), summaries of the Persons workplaces and places of study. |

**params**

| | | | |
|---|---|---|---|
| 1 | `personId` | ID | |
| 2 | `firstName` | String | |

**result**

| | | | | |
|---|---|---|---|---|
| 1 | `otherPerson.id` | ID | R | |
| 2 | `otherPerson.lastName` | String | R | |
| 3 | `distanceFromPerson` | 32-bit Integer | C | |
| 4 | `otherPerson.birthday` | Date | R | |
| 5 | `otherPerson.creationDate` | DateTime | R | |
| 6 | `otherPerson.gender` | String | R | |
| 7 | `otherPerson.browserUsed` | String | R | |
| 8 | `otherPerson.locationIP` | String | R | |
| 9 | `otherPerson.email` | {Long String} | R | |
| 10 | `otherPerson.speaks` | {String} | R | |
| 11 | `locationCity.name` | String | R | |
| 12 | `universities` | {<String, 32-bit Integer, String>} | A | `{‹university.name, studyAt.classYear, universityCity.name›}` |
| 13 | `companies` | {<String, 32-bit Integer, String>} | A | `{‹company.name, workAt.workFrom, companyCountry.name›}` |

**sort**

| | | | |
|---|---|---|---|
| 1 | `distanceFromPerson` | ↑ | |
| 2 | `otherPerson.lastName` | ↑ | |
| 3 | `otherPerson.id` | ↑ | |

| | |
|---|---|
| limit | 20 |
| CPs | 2.1, 5.3, 8.2 |
| relevance | This query is a representative of a simple navigational query. It looks for paths of length 1..3 through the knows relation, starting from a given Person and ending at a Person with a given first name. It is interesting for several aspects. (1) It requires for a complex aggregation for returning the concatenation of universities, companies, languages and email information of the Person. (2) It tests the ability of the optimizer to move the evaluation of sub-queries functionally dependant on the Person, after the evaluation of the top-k. (3) Its performance is highly sensitive to properly estimating the cardinalities in each transitive path, and paying attention not to explore already visited Persons. |

## Interactive / complex / 2

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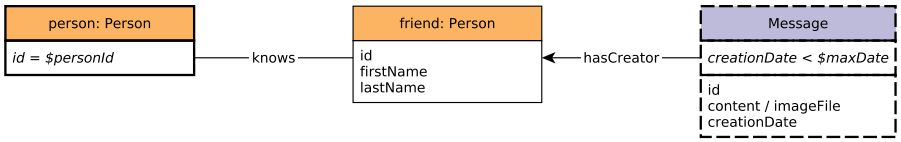
|---|---|
| query | Interactive / complex / 2 |
| title | Recent messages by your friends |
| pattern |  |
| desc. | Given a start Person (`person`), find the most recent Messages from all of that Person's friends (`friend` nodes). Only consider Messages created before the given `maxDate` (excluding that day). |

**params**

| | | | |
|---|---|---|---|
| 1 | `personId` | ID | |
| 2 | `maxDate` | Date | |

**result**

| | | | | |
|---|---|---|---|---|
| 1 | `friend.id` | ID | R | |
| 2 | `friend.firstName` | String | R | |
| 3 | `friend.lastName` | String | R | |
| 4 | `message.id` | ID | R | |
| 5 | `message.content` or `message.imageFile` (for photos) | Text | R | |
| 6 | `message.creationDate` | DateTime | R | |

**sort**

| | | | |
|---|---|---|---|
| 1 | `message.creationDate` | ↓ | |
| 2 | `message.id` | ↑ | |

| | |
|---|---|
| limit | 20 |
| CPs | 1.1, 2.2, 2.3, 3.2, 8.5 |
| relevance | This is a navigational query looking for paths of length two, starting from a given Person, going to their friends and from them, moving to their published Posts and Comments. This query exercices both the optimizer and how data is stored. It tests the ability to create execution plans taking advantage of the orderings induced by some operators to avoid performing expensive sorts. This query requires selecting Posts and Comments based on their creation date, which might be correlated with their identifier and therefore, having intermediate results with interesting orders. Also, messages could be stored in an order correlated with their creation date to improve data access locality. Finally, as many of the attributes required in the projection are not needed for the execution of the query, it is expected that the query optimizer will move the projection to the end. |

## Interactive / complex / 3

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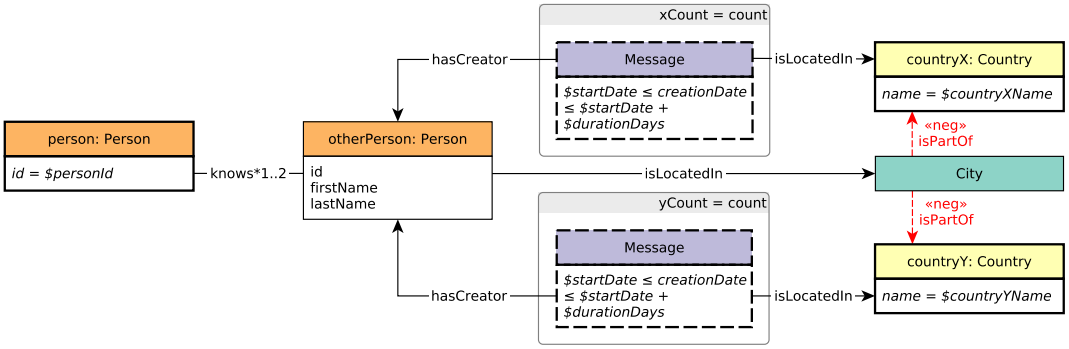
|---|---|
| query | Interactive / complex / 3 |
| title | Friends and friends of friends that have been to given countries |
| pattern |   |
| desc. | Given a start Person, find Persons that are their friends and friends of friends (excluding start Person) that have made Posts / Comments in both of the given Countries, `CountryX` and `CountryY`, within a given period. Only Persons that are foreign to Countries `CountryX` and `CountryY` are considered, that is Persons whose location is neither `CountryX` nor `CountryY`. |

params

| | | | |
|---|---|---|---|
| 1 | `personId` | ID | |
| 2 | `countryXName` | String | |
| 3 | `countryYName` | String | |
| 4 | `startDate` | Date | Beginning of requested period |
| 5 | `durationDays` | 32-bit Integer | Duration of requested period, in days. The interval `[startDate, startDate + durationDays)` is closed-open |

result

| | | | | |
|---|---|---|---|---|
| 1 | `otherPerson.id` | ID | R | |
| 2 | `otherPerson.firstName` | String | R | |
| 3 | `otherPerson.lastName` | String | R | |
| 4 | `xCount` | 32-bit Integer | A | Number of Messages from Country `CountryX` created by the Person within the given time |
| 5 | `yCount` | 32-bit Integer | A | Number of Messages from Country `CountryY` created by the Person within the given time |
| 6 | `count` | 32-bit Integer | A | `count` = `xCount` + `yCount` |

sort

| | | | |
|---|---|---|---|
| 1 | `xCount` | ↓ | |
| 2 | `otherPerson.id` | ↑ | |

| | |
|---|---|
| limit | 20 |
| CPs | 2.1, 3.1, 5.1, 8.2, 8.5 |
| relevance | This query looks for paths of length two and three, starting from a Person, going to friends or friends of friends, and then moving to Messages. This query tests the ability of the query optimizer to select the most efficient join ordering, which will depend on the cardinalities of the intermediate results. Many friends of friends can be duplicate, then it is expected to eliminate duplicates and those people prior to access the Post and Comments, as well as eliminate those friends from Countries `CountryX` and `CountryY`, as the size of the intermediate results can be severely affected. A possible structural optimization could be to materialize the number of Posts and Comments created by a Person, and progressively filter those people that could not even fall in the top 20 even having all their posts in the Countries `CountryX` and `CountryY`. |

## Interactive / complex / 4

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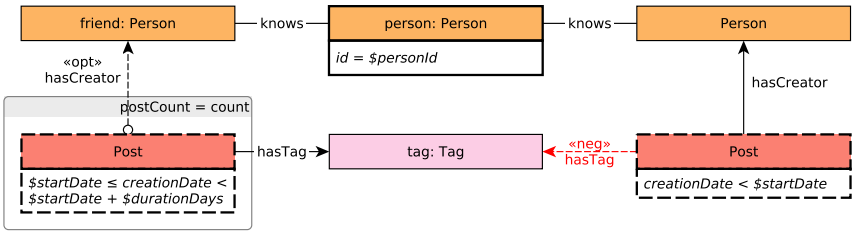
|---|---|
| query | Interactive / complex / 4 |
| title | New topics |
| pattern |  |
| desc. | Given a start Person (`personId`), find Tags that are attached to Posts that were created by that Person's friends. Only include Tags that were attached to friends' Posts created within a given time interval, and that were never attached to friends' Posts created before this interval. |

**params**

| | | | |
|---|---|---|---|
| 1 | `personId` | ID | |
| 2 | `startDate` | Date | |
| 3 | `durationDays` | 32-bit Integer | Duration of requested period, in days. The interval `[startDate, startDate + durationDays)` is closed-open |

**result**

| | | | | |
|---|---|---|---|---|
| 1 | `tag.name` | Long String | R | |
| 2 | `postCount` | 32-bit Integer | A | Number of Posts made within the given time interval that have `tag` |

**sort**

| | | |
|---|---|---|
| 1 | `postCount` | ↓ |
| 2 | `tag.name` | ↑ |

| | |
|---|---|
| limit | 10 |
| CPs | 2.3, 8.2, 8.5 |
| relevance | This query looks for paths of length two, starting from a given Person, moving to Posts and then to Tags. It tests the ability of the query optimizer to properly select the usage of hash joins or index based joins, depending on the cardinality of the intermediate results. These cardinalities are clearly affected by the input Person, the number of friends, the variety of Tags, the time interval and the number of Posts. |

## Interactive / complex / 5

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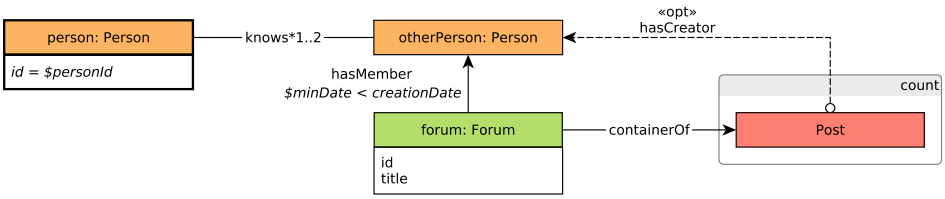
|---|---|
| query | Interactive / complex / 5 |
| title | New groups |
| pattern |  |
| desc. | Given a start Person, find the Forums which that Person's friends and friends of friends (excluding start Person) became Members of after a given date. For each Forum find the number of Posts that were created by any of these Persons. For each Forum and consider only those Persons which joined that particular Forum after the given date (`minDate`). |
| params | 1 `personId` ID<br>2 `minDate` Date |
| result | 1 `forum.title` Long String R<br>2 `postCount` 32-bit Integer A Number of Posts made in `forum` that were created by friends |
| sort | 1 `postCount` ↓<br>2 `forum.id` ↑ |
| limit | 20 |
| CPs | 2.3, 3.3, 8.2, 8.5 |
| relevance | This query looks for paths of length two and three, starting from a given Person, moving to friends and friends of friends, and then getting the Forums they are members of. Besides testing the ability of the query optimizer to select the proper join operator, it rewards the usage of indexes, but their accesses will be presumably scattered due to the two/three-hop search space of the query, leading to unpredictable and scattered index accesses. Having efficient implementations of such indexes will be highly beneficial. |

## Interactive / complex / 6

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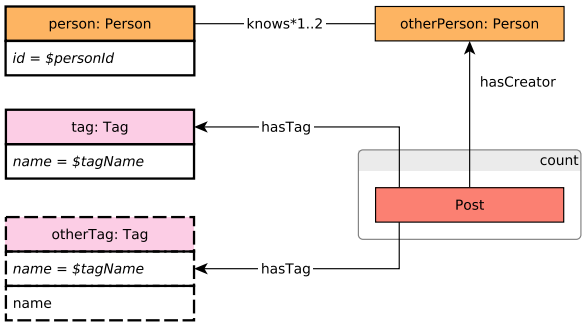
|---|---|
| query | Interactive / complex / 6 |
| title | Tag co-occurrence |
| pattern |  |
| desc. | Given a start Person and some Tag, find the other Tags that occur together with this Tag on Posts that were created by start Person's friends and friends of friends (excluding start Person). Return top 10 Tags, and the count of Posts that were created by these Persons, which contain both this Tag and the given Tag. |

**params**

| | | | |
|---|---|---|---|
| 1 | personId | ID | |
| 2 | tagName | Long String | |

**result**

| | | | | |
|---|---|---|---|---|
| 1 | otherTag.name | Long String | R | |
| 2 | postCount | 32-bit Integer | A | Number of Posts that were created by friends and friends of friends, which have the Tag otherTag |

**sort**

| | | | |
|---|---|---|---|
| 1 | postCount | ↓ | |
| 2 | otherTag.name | ↑ | |

| | |
|---|---|
| limit | 10 |
| CPs | 5.1, 8.2 |
| relevance | This query looks for paths of lengths three or four, starting from a given Person, moving to friends or friends of friends, then to Posts and finally ending at a given Tag. |

## Interactive / complex / 7

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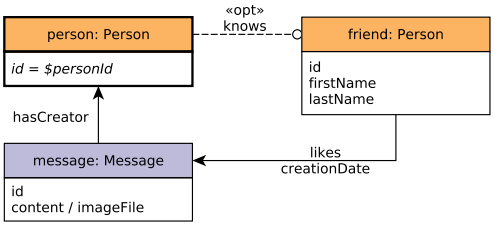
|---|---|
| query | Interactive / complex / 7 |
| title | Recent likers |
| pattern |  |
| desc. | Given a start Person, find (most recent) `likes` on any of start Person's Messages. Find Persons that liked (`likes` edge) any of start Person's Messages, the Messages they liked most recently, the creation date of that like, and the latency in minutes (`minutesLatency`) between creation of Messages and like. Additionally, for each Person found return a flag indicating (`isNew`) whether the liker is a friend of start Person. In case that a Person liked multiple Messages at the same time, return the Message with lowest identifier. |
| params | 1 `personId` ID |

result

| | | | | |
|---|---|---|---|---|
| 1 | `friend.id` | ID | R | |
| 2 | `friend.firstName` | String | R | |
| 3 | `friend.lastName` | String | R | |
| 4 | `likes.creationDate` | DateTime | R | |
| 5 | `message.id` | ID | R | |
| 6 | `message.content` or `message.imageFile` (for photos) | Text | R | |
| 7 | `minutesLatency` | 32-bit Integer | C | Duration between creation of the Message and the creation of the like, in minutes |
| 8 | `isNew` | Boolean | C | False if `person` and `friend` know each other, True otherwise |

sort

| | | |
|---|---|---|
| 1 | `likes.creationDate` | ↓ |
| 2 | `friend.id` | ↑ |

| | |
|---|---|
| limit | 20 |
| CPs | 2.2, 2.3, 3.3, 5.1, 8.1, 8.3 |
| relevance | This query looks for paths of length two, starting from a given Person, moving to its published messages and then to Persons who liked them. It tests several aspects related to join optimization, both at query optimization plan level and execution engine level. On the one hand, many of the columns needed for the projection are only needed in the last stages of the query, so the optimizer is expected to delay the projection until the end. This query implies accessing two-hop data, and as a consequence, index accesses are expected to be scattered. We expect to observe variate cardinalities, depending on the characteristics of the input parameter, so properly selecting the join operators will be crucial. This query has a lot of correlated sub-queries, so it is testing the ability to flatten the query execution plans. |

## Interactive / complex / 8

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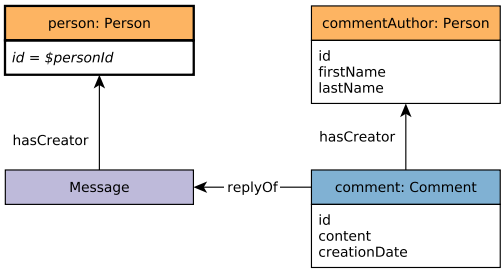
|---|---|
| query | Interactive / complex / 8 |
| title | Recent replies |
| pattern |  |
| desc. | Given a start Person, find (most recent) Comments that are replies to Messages of the start Person. Only consider direct (single-hop) replies, not the transitive (multi-hop) ones. Return the reply Comments, and the Person that created each reply Comment. |
| params | |

| | | | |
|---|---|---|---|
| 1 | personId | ID | |

result

| | | | | |
|---|---|---|---|---|
| 1 | commentAuthor.id | ID | R | |
| 2 | commentAuthor.firstName | String | R | |
| 3 | commentAuthor.lastName | String | R | |
| 4 | comment.creationDate | DateTime | R | |
| 5 | comment.id | ID | R | |
| 6 | comment.content | Text | R | |

sort

| | | | |
|---|---|---|---|
| 1 | comment.creationDate | ↓ | |
| 2 | comment.id | ↑ | |

| | |
|---|---|
| limit | 20 |
| CPs | 2.4, 3.3, 5.3 |
| relevance | This query looks for paths of length two, starting from a given Person, going through its created Messages and finishing at their replies. In this query there is temporal locality between the replies being accessed. Thus the top-k order by this can interact with the selection, i.e. do not consider older Posts than the 20th oldest seen so far. |

## Interactive / complex / 9

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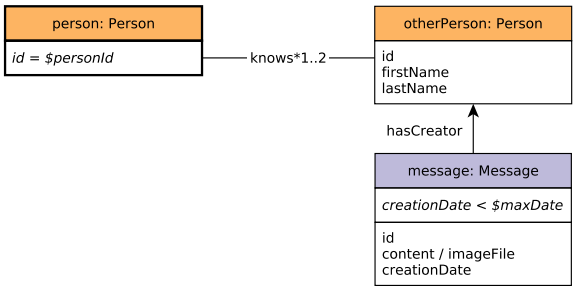
|---|---|
| query | Interactive / complex / 9 |
| title | Recent messages by friends or friends of friends |
| pattern |  |
| desc. | Given a start Person, find (the most recent) Messages created by that Person's friends or friends of friends (excluding start Person). Only consider Messages created before the given `maxDate` (excluding that day). |

**params**

| | | | |
|---|---|---|---|
| 1 | `personId` | ID | |
| 2 | `maxDate` | Date | |

**result**

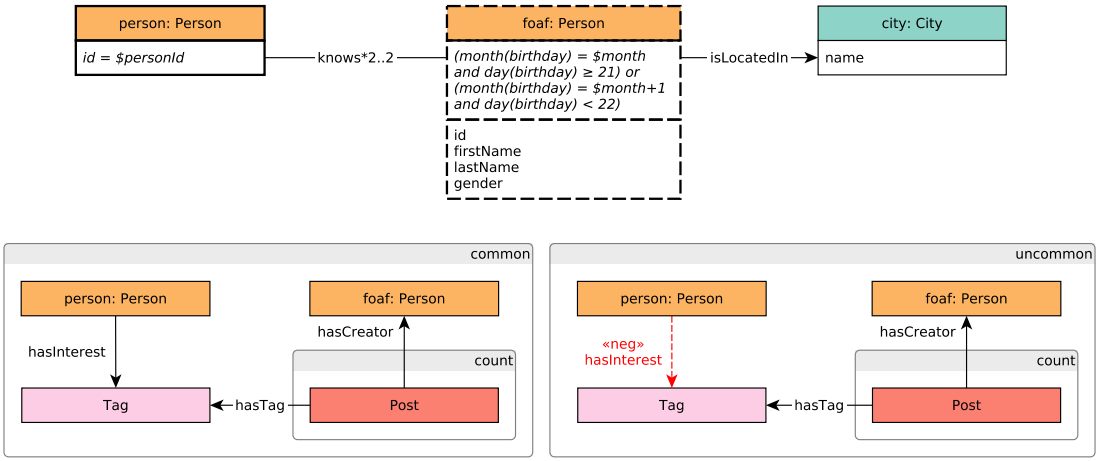
| | | | | |
|---|---|---|---|---|
| 1 | `otherPerson.id` | ID | R | |
| 2 | `otherPerson.firstName` | String | R | |
| 3 | `otherPerson.lastName` | String | R | |
| 4 | `message.id` | ID | R | |
| 5 | `message.content` or `message.imageFile` (for photos) | Text | R | |
| 6 | `message.creationDate` | DateTime | R | |

**sort**

| | | | |
|---|---|---|---|
| 1 | `message.creationDate` | ↓ | |
| 2 | `message.id` | ↑ | |

| | |
|---|---|
| limit | 20 |
| CPs | 1.1, 1.2, 2.2, 2.3, 3.2, 3.3, 8.5 |
| relevance | This query looks for paths of length two or three, starting from a given Person, moving to its friends and friends of friends, and ending at their created Messages. This is one of the most complex queries, as the list of choke points indicates. This query is expected to touch variable amounts of data with entities of different characteristics, and therefore, properly estimating cardinalities and selecting the proper operators will be crucial. |

## Interactive / complex / 10

| | |
|---|---|
| query | Interactive / complex / 10 |
| title | Friend recommendation |
| pattern |  |
| desc. | Given a start Person with id `personId`, find that Person's friends of friends (`foaf`) – excluding the start Person and his/her immediate friends –, who were born on or after the 21st of a given `month` (in any year) and before the 22nd of the following month. Calculate the similarity between each `friend` and the start `person`, where `commonInterestScore` is defined as follows:<br>• `common` = number of Posts created by `friend`, such that the Post has a Tag that the start `person` is interested in<br>• `uncommon` = number of Posts created by `friend`, such that the Post has no Tag that the start `person` is interested in<br>• `commonInterestScore` = `common` - `uncommon` |



**params**

| | | | |
|---|---|---|---|
| 1 | `personId` | ID | |
| 2 | `month` | 32-bit Integer | Between 1 and 12. Implementations may also pass the next month as an additional `nextMonth` parameter |

**result**

| | | | | |
|---|---|---|---|---|
| 1 | `foaf.id` | ID | R | |
| 2 | `foaf.firstName` | String | R | |
| 3 | `foaf.lastName` | String | R | |
| 4 | `commonInterestScore` | 32-bit Integer | A | |
| 5 | `foaf.gender` | String | R | |
| 6 | `city.name` | String | R | |

**sort**

| | | | |
|---|---|---|---|
| 1 | `commonInterestScore` | ↓ | |
| 2 | `foaf.id` | ↑ | |

| | |
|---|---|
| limit | 10 |
| CPs | 2.3, 3.3, 4.1, 4.2, 5.1, 5.2, 6.1, 7.1, 8.6 |
| relevance | This query looks for paths of length two, starting from a Person and ending at the friends of their friends. It does widely scattered graph traversal, and one expects no locality of in friends of friends, as these have been acquired over a long time and have widely scattered identifiers. The join order is simple but one must see that the anti-join for "not in my friends" is better with hash. Also the last pattern in the scalar sub-queries joining or anti-joining the Tags of the candidate's Posts to interests of self should be by hash. |

## Interactive / complex / 11

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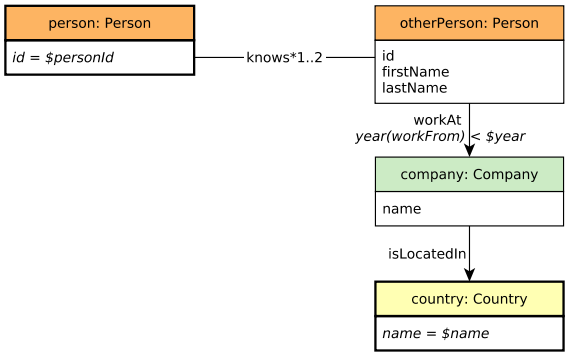
|---|---|
| query | Interactive / complex / 11 |
| title | Job referral |
| pattern |  |
| desc. | Given a start Person, find that Person's friends and friends of friends (excluding start Person) who started working in some Company in a given Country, before a given date (year). |
| params | |
| result | |
| sort | |
| limit | 10 |
| CPs | 1.3, 2.3, 2.4, 3.3, 4.2 |
| relevance | This query looks for paths of length two or three, starting from a Person, moving to friends or friends of friends, and ending at a Company. In this query, there are selective joins and a top-k order by that can be exploited for optimizations. |

| | | | |
|---|---|---|---|
| 1 | personId | ID | |
| 2 | countryName | String | |
| 3 | workFromYear | 32-bit Integer | |

| | | | | |
|---|---|---|---|---|
| 1 | otherPerson.id | ID | R | |
| 2 | otherPerson.firstName | String | R | |
| 3 | otherPerson.lastName | String | R | |
| 4 | company.name | String | R | |
| 5 | workAt.workFrom | 32-bit Integer | R | |

| | | | |
|---|---|---|---|
| 1 | workAt.workFrom | ↑ | |
| 2 | otherPerson.id | ↑ | |
| 3 | company.name | ↓ | |

## Interactive / complex / 12

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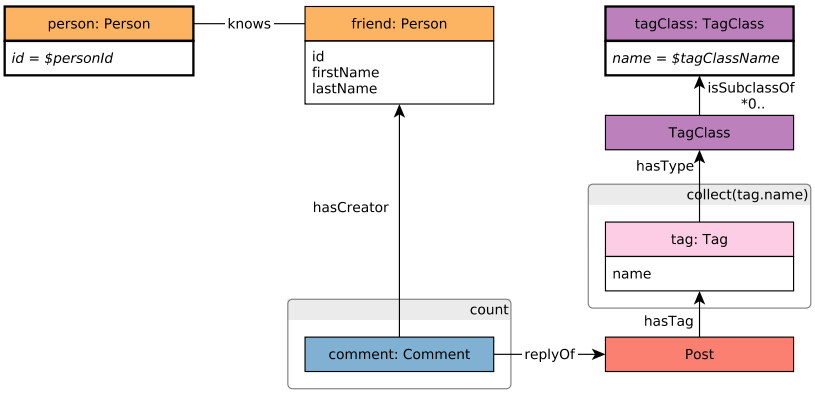
|---|---|
| query | Interactive / complex / 12 |
| title | Expert search |
| pattern |  |
| desc. | Given a start Person, find the Comments that this Person's friends made in reply to Posts, considering only those Comments that are direct (single-hop) replies to Posts, not the transitive (multi-hop) ones. Only consider Posts with a Tag in a given TagClass or in a descendent of that TagClass. Count the number of these reply Comments, and collect the Tags that were attached to the Posts they replied to, but only collect Tags with the given TagClass or with a descendant of that TagClass. Return Persons with at least one reply, the reply count, and the collection of Tags. |

**params**

| | | | |
|---|---|---|---|
| 1 | personId | ID | |
| 2 | tagClassName | Long String | |

**result**

| | | | | |
|---|---|---|---|---|
| 1 | friend.id | ID | R | |
| 2 | friend.firstName | String | R | |
| 3 | friend.lastName | String | R | |
| 4 | tagNames | {Long String} | A | |
| 5 | replyCount | 32-bit Integer | A | |

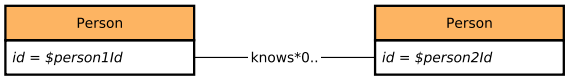
**sort**

| | | | |
|---|---|---|---|
| 1 | replyCount | ↓ | |
| 2 | friend.id | ↑ | |

| | |
|---|---|
| limit | 20 |
| CPs | 3.3, 7.2, 7.3, 8.2 |
| relevance | This query starts at a Person, moves to its friends, and the to their Comments and their root Posts. Then, it gets the Tag of each Post and checks whether it (directly or transitively) belongs to the specified TagClass. This can be thought of a bidirectional search between the Person and the TagClass. The difficulty of this query is determining the optimal direction of this traversal. |

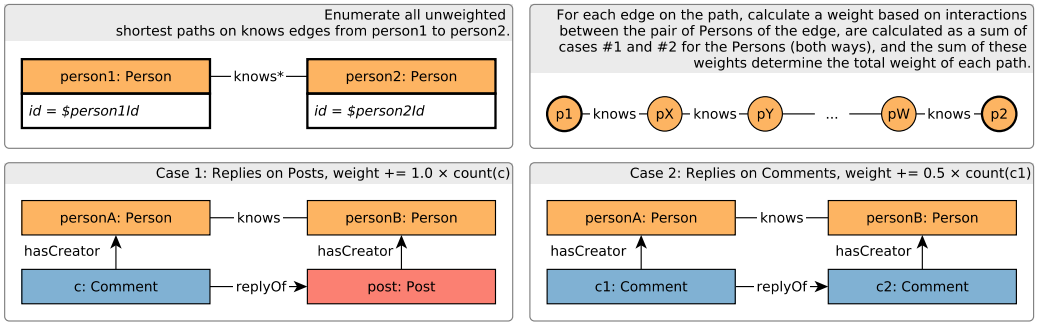
## Interactive / complex / 13



| | |
|---|---|
| query | Interactive / complex / 13 |
| title | Single shortest path |
| pattern |  |
| desc. | Given two Persons, find the shortest path between these two Persons in the subgraph induced by the knows relationships. Return the length of this path:<br>• −1: no path found<br>• 0: start person = end person<br>• > 0: path found (start person ≠ end person) |
| params | 1 `person1.id` ID `person1Id`<br>2 `person2.id` ID `person2Id` |
| result | 1 `shortestPathLength` 32-bit Integer `C` |
| CPs | 3.3, 7.2, 7.3, 7.5, 8.1, 8.6 |
| relevance | This query looks for a variable length path, starting at a given Person and finishing at an another given Person. Proper cardinality estimation and search space prunning, will be crucial. This query also allows for possible parallel implementations. |

## Interactive / complex / 14

IC 1
IC 2
IC 3
IC 4
IC 5
IC 6
IC 7
IC 8
IC 9
IC 10
IC 11
IC 12
IC 13
IC 14

| | |
|---|---|
| query | Interactive / complex / 14 |
| title | Trusted connection paths |
| pattern |   |
| desc. | Given two Persons, find all (unweighted) shortest paths between these two Persons, in the subgraph induced by the knows relationship.<br>Then, for each path calculate a weight. The nodes in the path are Persons, and the weight of a path is the sum of weights between every pair of consecutive Person nodes in the path.<br>The weight for a pair of Persons is calculated based on their interactions:<br>• Every direct reply (by one of the Persons) to a Post (by the other Person) contributes 1.0.<br>• Every direct reply (by one of the Persons) to a Comment (by the other Person) contributes 0.5.<br>Note that interactions are counted both ways (e.g. if Alice writes 2 Post replies and 1 Comment reply to Bob, while Bob writes 3 Post replies and 4 Comment replies to Alice, their interaction score is $2 \times 1.0 + 1 \times 0.5 + 3 \times 1.0 + 4 \times 0.5 = 7.5$).<br>Return all the paths with shortest length, and their weights. Do not return any rows if there is no path between the two Persons. |
| params | 1 person1Id ID<br>2 person2Id ID |
| result | 1 personIdsInPath [ID] C identifiers representing an ordered sequence of the Persons in the path<br>2 pathWeight 64-bit Float C |
| sort | 1 pathWeight ↓ The order of paths with the same weight is unspecified |
| CPs | 3.3, 5.3, 7.2, 7.3, 7.5, 7.7, 8.1, 8.2, 8.3, 8.6 |
| relevance | This query looks for a variable length path, starting at a given Person and finishing at an another given Person. This is a more complex query as it not only requires computing the path length, but returning it and computing a weight. To compute this weight one must look for smaller sub-queries with paths of length three, formed by the two Persons at each step, a Post and a Comment. |